

Introduction to Biophysics II: Quantum physics & biology

Lecture 15: Electron transfer in biology: 2

Reading:

- Quantum Effects in Biology, chapter 9
- Bolton et al.; Electron Transfer in Inorganic, Organic, and Biological Systems. Advances in Chemistry; American Chemical Society: Washington, DC, 1991
- Newton MD, Chem. Rev.91, 767, 1991
- Marcus RA, Sutin N. BBA 811 (1985) 265
- Marcus RA, JCP 24 (1956) 966
- C.C. Page, C.C. Moser, X.X. Chen, P.L. Dutton, Natural engineering principles of electron tunneling in biological oxidation-reduction, Nature 402 (1999) 47–52

1

Electron transfer: semiclassical

Previously on Biophys 2

$$k_{ET} \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta E_b}\right) \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

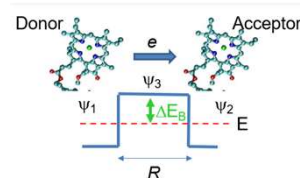
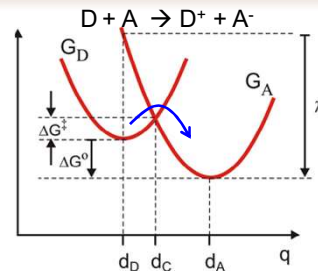
$$\exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

- classical: crossing energy barrier $\Delta G^\ddagger$
= Boltzmann $\exp(-\Delta G^\ddagger/kT)$

$$\Delta G^\ddagger = \frac{1}{4\lambda}[\Delta G^\circ + \lambda]^2$$

$$\exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta E_b}\right)$$

- Quantum-mechanical tunneling
through a barrier
 ΔE_b = barrier height,
R = barrier width

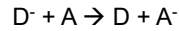


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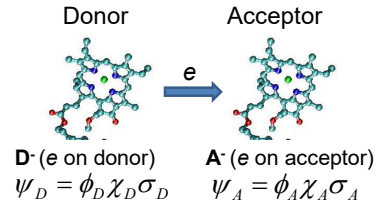
Electron transfer: Quantum

Quantum-mechanical approach

Transfer of electron from reduced donor to neutral acceptor:



Consider wave function of electron on donor (initial state) and on acceptor (final):
(Born-Oppenheimer approximation)



Construct superposition state for the electron:

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

ϕ - spatial wave function
 χ - nuclear wave function
 σ - spin

Initial state: $C_D=1, C_A=0$.

Need to solve time-dependent Schrödinger equation to find $C_A(t)$: $\hat{H}\Psi = i\hbar \frac{\partial \Psi}{\partial t}$

The solution is mathematically similar to Förster/Dexter energy transfer:

Recall
Lecture 6

$$C_2(t) = -\frac{i}{\hbar} \tilde{H}_{21} \frac{\exp(i\omega_0 t) - 1}{i\omega_0} \quad \omega_0 \equiv (E_2 - E_1) / \hbar$$

C_A

$\langle \psi_2 | \tilde{H}_{21} | \psi_1 \rangle$

$\leftarrow \langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle$

$E_1 = E_D$
 $E_2 = E_A$

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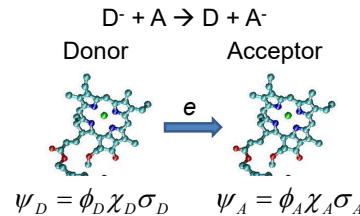
Electron transfer

Quantum-mechanical approach

$$\Psi = C_D \phi_D \chi_D \sigma_D + C_A \phi_A \chi_A \sigma_A$$

Initial state: $C_D=1, C_A=0$:

$$\langle \phi_A \chi_A \sigma_A | H_{AD} | \phi_D \chi_D \sigma_D \rangle$$



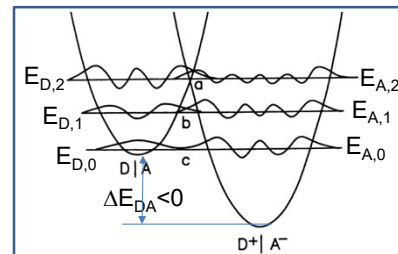
Solution:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} \left| \langle \chi_{A,m} | \chi_{D,n} \rangle \right|^2 \times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \times \langle \sigma_A | \sigma_D \rangle \right]$$

Energy in vib. state m of acceptor
Energy in vib. state n of donor

Recall Lecture 7:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(el)}|^2 \sum_{n,m,u,w} \left[\frac{\exp(-E_{n(1)}/kT)}{Z_1} \left| \langle \chi_{1m} | \chi_{1n} \rangle \right|^2 \times \frac{\exp(-E_{u(2)}/kT)}{Z_2} \left| \langle \chi_{2w} | \chi_{2u} \rangle \right|^2 \times \delta(\Delta E_{1nm} - \Delta E_{2uw}) \right]$$



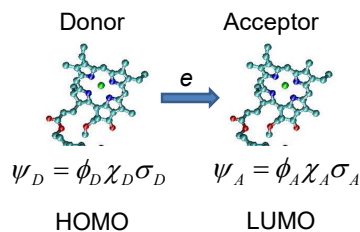
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Electron transfer

Quantum-mechanical approach

Solution:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \times \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \right] \times \times \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \times \langle \sigma_A | \sigma_D \rangle$$



1. Interaction

Recall: Born-Oppenheimer approximation, electron wavefunction separates from nuclear:

$$\begin{aligned} \langle \phi_A \chi_A \sigma_A | \tilde{H}_{AD} | \phi_D \chi_D \sigma_D \rangle &= \langle \phi_A(\vec{r}) \chi_A(\vec{R}) \sigma_A(s) | \tilde{H}_{AD} | \phi_D(\vec{r}) \chi_D(\vec{R}) \sigma_D(s) \rangle \\ &= \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle \langle \chi_A | \chi_D \rangle \langle \sigma_A | \sigma_D \rangle \end{aligned}$$

$$H_{AD(el)} \equiv \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle$$

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Electron transfer

Quantum-mechanical approach

Electron interaction:

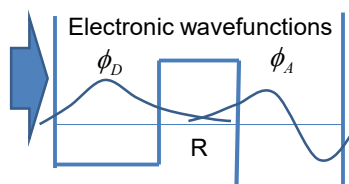
$$H_{AD(el)} \equiv \langle \phi_A | \tilde{H}_{AD} | \phi_D \rangle$$

In the simplest form, this is just electron tunneling:

$$H_{AD} \propto \sqrt{T} \propto \exp \left[- \left(\frac{\sqrt{2m_e \Delta E_b}}{\hbar} \right) R_{DA} \right]$$

For constant interaction:

$$H_{DA(el)} \propto \langle \phi_A | \phi_D \rangle$$



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Electron transfer

Quantum-mechanical approach

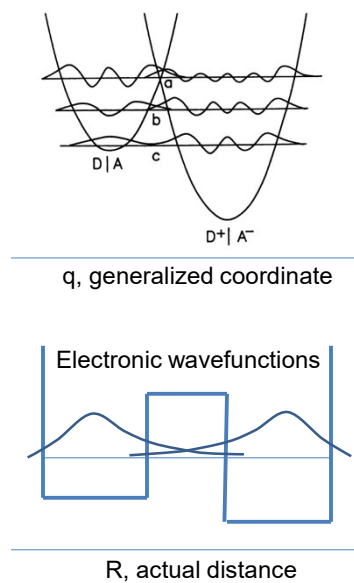
Overlap of nuclear

$$\langle \chi_{A,m} | \chi_{D,n} \rangle$$

and

electronic wavefunctions

$$H_{DA(el)} \propto \langle \phi_A | \phi_D \rangle$$

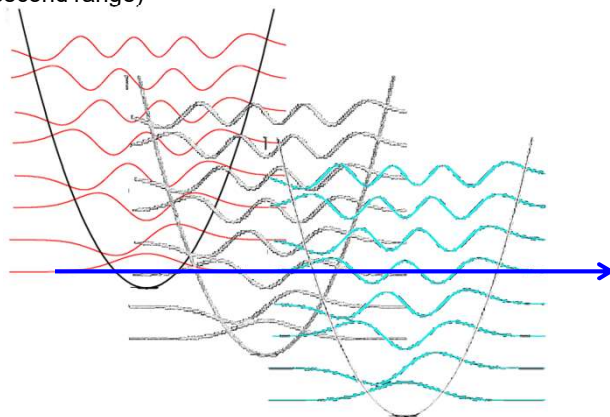


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Electron transfer

Quantum-mechanical approach: fast multistep ET

Note: When electron transfer is fast, there may be no time for each state to relax to the lowest vibrational state $\rightarrow$ further transfer can occur at locally higher temperatures with better efficiency (typically must be in the sub-picosecond or sub-10-picosecond range)



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Electron transfer

Split Franck-Condon factors into solvent/matrix and internal vibrational parts:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \delta(E_{A,m} - E_{D,n} + \Delta E_{DA}) \right]$$

Omit spins (but they are here)

$$\downarrow$$

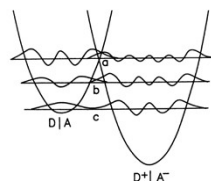
$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \times \exp\left(-\frac{[\lambda_{bath} + E_{A,m} - E_{D,n} + \Delta E_{DA}]^2}{4\lambda_{bath}kT}\right) \right]$$

Where λ_{bath} is the reorganization energy of bath (surrounding solvent, protein)

The Franck-Condon integrals are now only for a limited number of internal vibrations.

→ Replace δ -function with a Gaussian distribution of probabilities due to the continuous vibrational spectrum of the surrounding protein (bath)

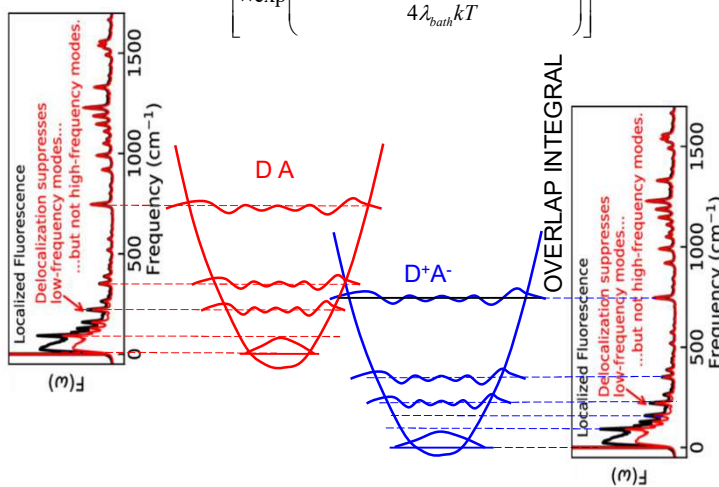
Gaussian (Normal) function
Maximum shifted by λ_{bath}



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Electron transfer

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\langle \chi_{A,m} | \chi_{D,n} \rangle|^2 \times \exp\left(-\frac{[\lambda_{bath} + E_{A,m} - E_{D,n} + \Delta E_{DA}]^2}{4\lambda_{bath}kT}\right) \right]$$



Mike Reppert: Phys. Chem. B 2020, 124, 45, 10024–10033 (log-normal is used instead of shifted Gaussian)

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Electron transfer

Split Franck-Condon factors

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} |\chi_{A,m} | \chi_{D,n}|^2 \times \exp\left(-\frac{[\lambda_{bath} + E_{A,m} - E_{D,n} + \Delta E_{DA}]^2}{4\lambda_{bath}kT}\right) \right]$$

As a crude approximation, the relevant high-frequency internal vibrations can be replaced with one vibrational mode at an averaged frequency ν :

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_m \left[\left(\frac{e^{-S} S^m}{m!} \right) \times \exp\left(-\frac{[\lambda_{bath} + \Delta E_{DA} + m\hbar\nu]^2}{4\lambda_{bath}kT}\right) \right]$$

Where the Huang-Rhys factor is $S = \frac{\lambda_{internal}}{\hbar\nu}$ (see lecture 6)

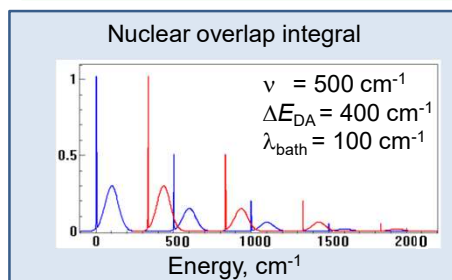
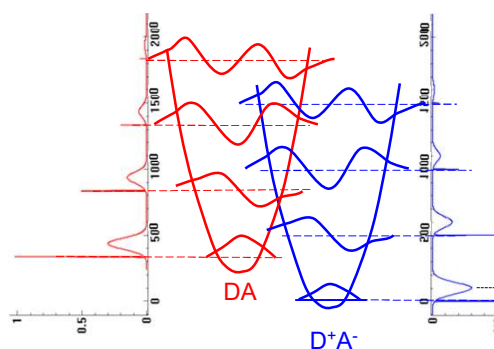
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Electron transfer

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_m \left[\left(\frac{e^{-S} S^m}{m!} \right) \times \exp\left(-\frac{[\lambda_{bath} + \Delta E_{DA} + m\hbar\nu]^2}{4\lambda_{bath}kT}\right) \right]$$

Franck-Condon series

"Gaussian" broadening due to the bath



Note: illustration only
 ΔE_{DA} is usually much larger

λ_{bath}

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Electron transfer

Split Franck-Condon factors into solvent/matrix and internal vibrational parts:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \sum_m \left[\left(\frac{e^{-S} S^m}{m!} \right) \times \exp \left(\frac{[\lambda_{bath} + \Delta E_{DA} + m\hbar\nu]^2}{4\lambda_{bath}kT} \right) \right]$$

$$S = \left| \langle \chi_{A,m} | \chi_{D,0} \rangle \right|^2 = \frac{\lambda_{internal}}{\hbar\nu}$$

It maximizes at $-\Delta G_0 = -\Delta E_{DA} \sim \lambda_{bath} + \lambda_{internal} = \lambda$

For $-\Delta G_0 < \lambda$ it overlaps well with classical expression:

$$k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{bath}kT)^{-1/2} \exp \left(\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT} \right)$$

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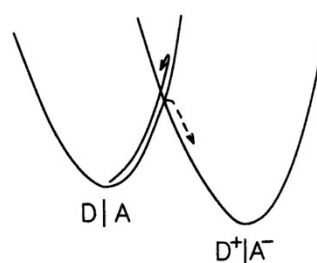
Electron transfer

Quantum-mechanical approach:

Weak coupling (nonadiabatic)

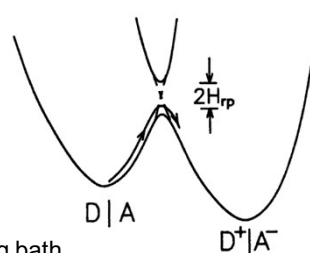
Previous derivation:

- Perturbation theory only works with weak coupling.
- Electron states on the donor and acceptor do not mix; the electron is localized at all times



Strong coupling (adiabatic)

- Perturbation theory does not work
- The approach is similar to the *exciton* theory, in which there will be new stationary states with an electron shared between two molecules:



Adiabatic = no heat transfer to/from the surrounding bath

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Electron transfer

Quantum-mechanical approach:

Strong coupling (adiabatic):

$$\psi = C_D \psi_D + C_A \psi_A = \begin{pmatrix} C_D \\ C_A \end{pmatrix}$$

$$\mathbf{H} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}$$

Solve: $\mathbf{H}\psi_B = E_B\psi_B$

Find new stationary states:

$$\begin{cases} \psi_+ = \sqrt{\frac{1+s}{2}}\psi_D + \sqrt{\frac{1-s}{2}}\psi_A \\ \psi_- = \sqrt{\frac{1-s}{2}}\psi_D - \sqrt{\frac{1+s}{2}}\psi_A \end{cases}$$

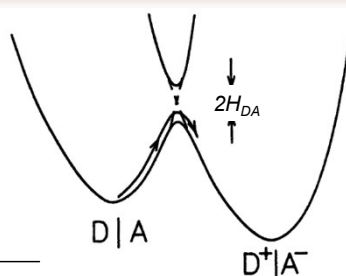
$$E_{\pm} = E_0 \pm \frac{1}{2}\sqrt{\delta^2 + 4H_{DA}^2}$$

$$E_0 = (H_{11} + H_{22}) / 2$$

$$\delta = H_{11} - H_{22}$$

$$H_{11} = E_D(Q)$$

$$H_{22} = E_A(Q)$$



*Near the crossing point, energies split!
Electrons are spreading over both molecules*

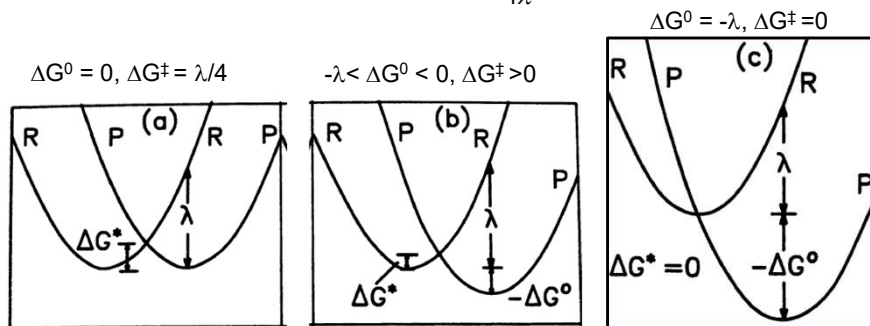
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Electron transfer

Different scenarios for ΔG^0 :

$$\ln k_{ET} = \text{const} - \left(\frac{2\sqrt{2m_e\Delta E_b}}{\hbar} \right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}$$

$$\Delta E_b \equiv \Delta G^\ddagger = \frac{1}{4\lambda} (\Delta G^0 - \lambda)^2$$



Fastest transfer is expected when $\Delta G^0 = -\lambda$

$D \rightarrow A$
=
 $R \rightarrow P$ (reactant $\rightarrow$ product)

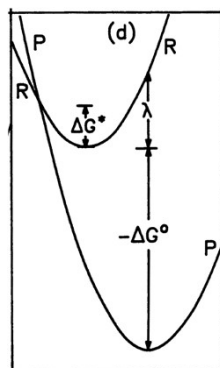
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Inverted Marcus Region

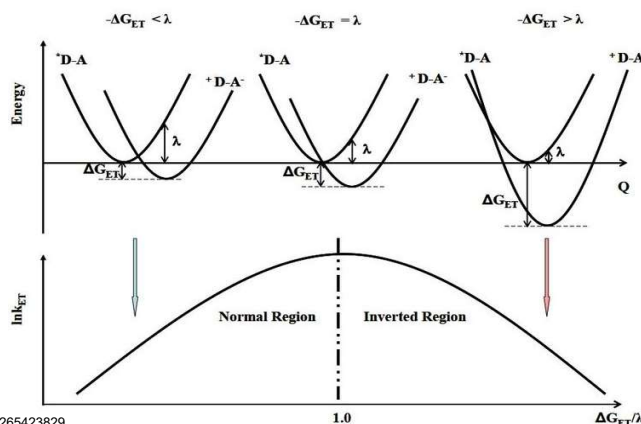
Different scenarios for ΔG^0 :

$$\Delta G^0 < -\lambda, \Delta E_b > 0$$

$$\ln k_{ET} = \text{const} - \left(\frac{2\sqrt{2m_e \Delta \bar{E}_b}}{\hbar} \right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}$$



Marcus: When acceptor energy becomes larger than λ , electron transfer rates start to decrease.
Rate vs ΔG^0 is a *symmetric* quadratic function around λ .



: <https://www.researchgate.net/publication/265423829>

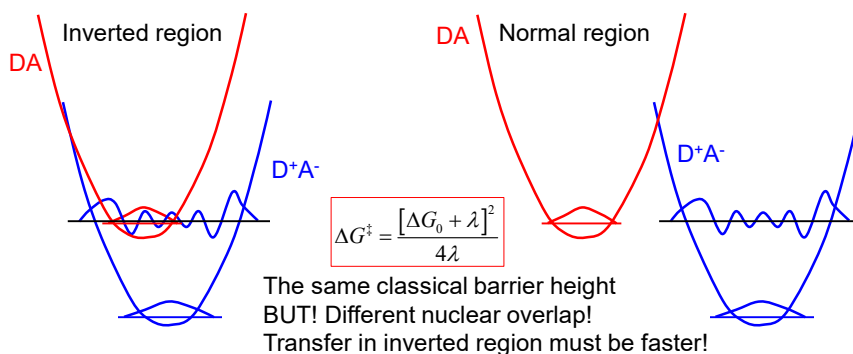
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Inverted Marcus Region

Different scenarios for ΔG^0 :

$$\Delta G^0 < -\lambda, \Delta G^\ddagger > 0$$

$$\ln k_{ET} = \text{const} - \left(\frac{2\sqrt{2m_e \Delta G_{average}^\ddagger}}{\hbar} \right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}$$



$$k_{rt} = \frac{2\pi}{\hbar} |H_{21(ef)}|^2 \sum_{n,m} \left[\frac{\exp(-E_{D,n}/kT)}{Z_1} \left| \langle \chi_{A,m} | \chi_{D,n} \rangle \right|^2 \times \delta(\Delta E_{DA} + E_{D,n} - E_{A,m}) \right] \times \langle \sigma_A | \sigma_D \rangle$$

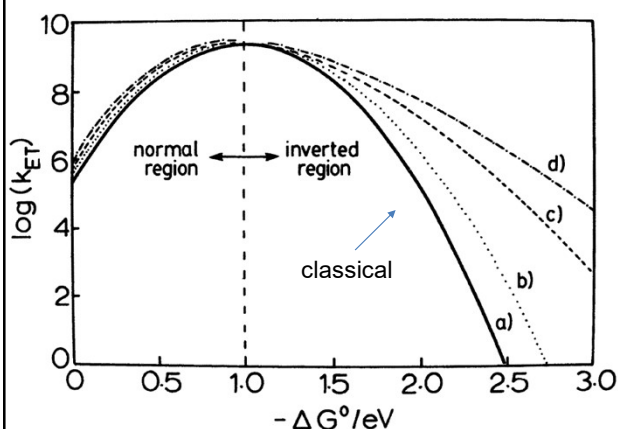
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Inverted Marcus Region

$$\text{Classical: } k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{\text{bath}}kT)^{-1/2} \exp\left(-\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}\right)$$

$$\text{Semi-classical: } k_{rt} = \frac{2\pi}{\hbar} |H_{AD(el)}|^2 (4\pi\lambda_{\text{bath}}kT)^{-1/2} \sum_m \left[\frac{e^{-S} S^m}{m!} \right] \times \exp\left(-\frac{[\lambda_{\text{bath}} + \Delta E_{DA} + m\hbar\nu]^2}{4\lambda_{\text{bath}}kT}\right)$$

$$S = \frac{\lambda_{\text{internal}}}{\hbar\nu}$$



Classical:

a) (symmetric)

Semi-classical using

vibration frequencies ν :

b) 750 cm^{-1} ,

c) 1500 cm^{-1} ,

d) 2250 cm^{-1} .

Other parameters:

$\lambda_{\text{bath}} = 0.8\text{ eV}$

$\lambda_{\text{internal}} = 0.2\text{ eV}$

$H_{21} = 0.3\text{ meV}$

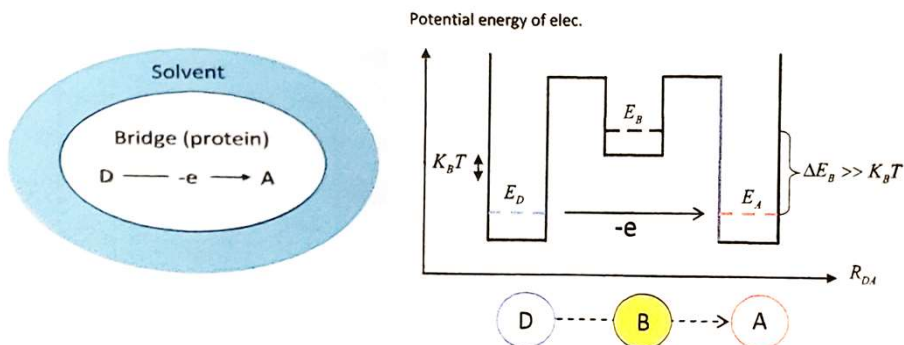
$T = 300\text{ K}$

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Electron transfer

Complication: inter-pigment space

It is not an empty protein environment!



The reaction coordinate diagram is not just two parabolas.

It may involve electronic states on other molecules (protein) in between.

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Electron transfer

Complication: inter-pigment space

It is not empty, protein environment

$$\text{Hamiltonian } \tilde{H} = \tilde{H}^{DA} + \tilde{H}^{br} + \tilde{V}$$

Where:

Donor and acceptor:

$$\tilde{H}^{DA} = \tilde{H}_D + \tilde{H}_A + \tilde{H}_{DA} \approx 0, \text{ too far!}$$

Where H_D and H_A are at Q where their energies cross at E_{res}

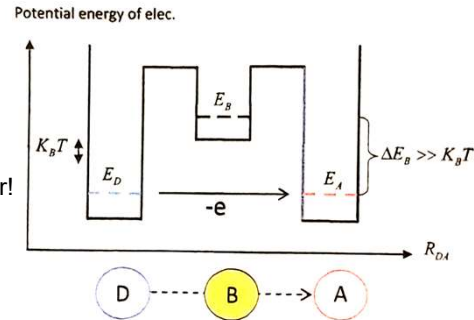
N bridges:

$$\tilde{H}^{br} = \sum_{i=1,N} \tilde{H}_{ii}^{br} + \sum_{i \neq j=1,N} \tilde{H}_{ij}^{br}$$

Interaction of D and A with bridges:

$$\tilde{V} = \sum_{i=1,N} (\tilde{V}_{D,bi} + \tilde{V}_{A,bi})$$

Electron wave functions for bridges are usually chosen to be localized on bridges



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Electron transfer

Complication: inter-pigment space

It is not empty, protein environment

Can have multiple path's

The tunneling matrix element:

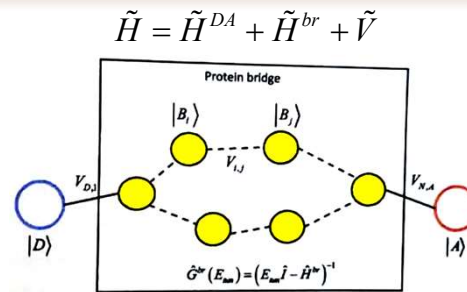
$$T_{DA} = \sum_{i=1\dots} \text{Path}_i$$

And for each path:

$$\text{Path}_i(D, B_{i1}, B_{i2}, \dots, B_{iN}, A) = V_{D,B_{i1}} \frac{1}{\Delta E_{i1}} V_{B_{i1}, B_{i2}} \frac{1}{\Delta E_{i2}} V_{B_{i2}, B_{i3}} \dots \frac{1}{\Delta E_{iN}} V_{B_{iN}, A}$$

$$\Delta E_{ik} = E_{res} - E_{B_{ik}}$$

Matrix element for each step is proportional to the coupling V and inversely proportional to the barrier height ΔE



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Electron transfer: Moser-Dutton

**Moser-Dutton semi-empirical electron transfer rate
(more practical for estimates)**

Had earlier:

$$\ln k_{ET} = \text{const} - \left(\frac{2\sqrt{2m_e\Delta\bar{E}_b}}{\hbar} \right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT}$$

Rewrite:

$$\log_{10} k_{ET} = A - \left(\frac{2\sqrt{2m_e\Delta\bar{E}_b}}{\hbar \ln 10} \right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT \ln 10}$$

Introduce:

$$\beta \equiv 2\hbar^{-1}\sqrt{2m_e\Delta\bar{E}_b}$$

$$\gamma \equiv \frac{1}{4kT \ln 10}$$

$$\log_{10} k_{ET} = A - \frac{\beta}{\ln 10} R_{DA} - \gamma \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$

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Electron transfer: Moser-Dutton

Moser-Dutton semi-empirical electron transfer rate

$$\log_{10} k_{ET} = A - \frac{\beta}{\ln 10} R_{DA} - \gamma \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$

Constant: γ

$$\gamma \equiv \frac{1}{4kT \ln 10} = 4.23 eV^{-1}$$

The respective term assumes that electron transfer occurs only via the cross-section of nuclear vibrational parabolas (classical barrier)
However, with quantum nuclear overlap, it can occur faster at energies below $\Delta G^\ddagger$.

Constant: β

Recall: $T \propto \exp\left(-\frac{2R}{\hbar}\sqrt{2m\Delta\bar{E}_b}\right) = \exp(-\beta R_{DA})$

β is a function of barrier height and R_{DA} , edge-to-edge distance
hard to calculate, it depends on inter-pigment media

Semi-empirical solution: adjust A, β , and γ by fitting experimental data

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Electron transfer: Moser-Dutton

Moser-Dutton semi-empirical electron transfer rate:

$$\log_{10} k_{ET} \approx 13 - 0.6 \left[\text{\AA}^{-1} \right] \left(R_{DA} - 3.6 \left[\text{\AA} \right] \right) - 3.1 \left[eV^{-1} \right] \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$

Note: this equation accounts for protein *empirically*, by scaling the terms of this equation derived from a simplified semi-classical model. Note that γ is smaller than its classical value of $4.23 eV^{-1}$, i.e. it is scaled by 0.735.

$\beta \approx (1.4 \text{ \AA})^{-1} = 0.6 \text{ \AA}^{-1}$ means that electron transfer time becomes 10 times slower when separation R_{DA} increases by $1.4 \text{ \AA}$

Deviations are expected, especially for closely spaced DA pairs...

NOTE: The equation works for $\Delta G_0 < 0$ (downhill) *normal* region:

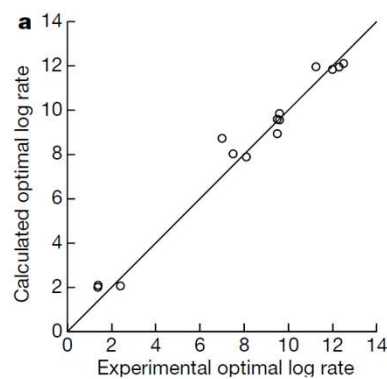
- C.C. Page, C.C. Moser, X.X. Chen, P.L. Dutton, Natural engineering principles of electron tunnelling in biological oxidation-reduction, Nature 402 (1999) 47–52

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Electron transfer: Moser-Dutton

Moser-Dutton semi-empirical electron transfer rate:

$$\log_{10} k_{ET} \approx 13 - 0.6 \left[\text{\AA}^{-1} \right] \left(R_{DA} - 3.6 \left[\text{\AA} \right] \right) - 3.1 \left[eV^{-1} \right] \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$



Comparison of calculated with experimental electron-transfer rates for the photosynthetic reaction centers of *Rhodospirillum rubrum* and *Rp. viridis*. The standard deviation is 0.5 log units.

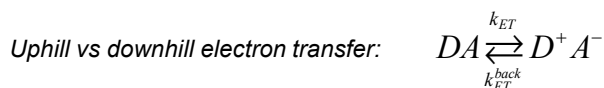
- C.C. Page, C.C. Moser, X.X. Chen, P.L. Dutton, Natural engineering principles of electron tunnelling in biological oxidation-reduction, Nature 402 (1999) 47–52

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Electron transfer: Moser-Dutton

Moser-Dutton ET rate was derived for downhill transfers, $\Delta G_0 < 0$:

$$\log_{10} k_{ET} \approx 13 - 0.6 \left[\text{\AA}^{-1} \right] \left(R_{DA} - 3.6 \left[\text{\AA} \right] \right) - 3.1 \left[eV^{-1} \right] \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$



Thermodynamics specifies that DA and D^+A^- state populations should follow Boltzmann statistics: $\frac{N_{DA}}{N_{D^+A^-}} = \exp\left(\frac{\Delta G_0}{kT}\right)$ Recall that $\Delta G_0 = G_{D^+A^-} - G_{DA}$ (< 0 for downhill transfer)

On the other hand, we have rate equation: $\frac{dN_{DA}}{dt} = -k_{ET} N_{DA} + k_{ET}^{back} N_{D^+A^-}$

In dynamic equilibrium: $0 = \frac{dN_{DA}}{dt} = -k_{ET} N_{DA} + k_{ET}^{back} N_{D^+A^-}$

$$k_{ET}^{back} = \frac{N_{DA}}{N_{D^+A^-}} k_{ET} = k_{ET} \exp\left(\frac{\Delta G_0}{kT}\right)$$

Note: laws of thermodynamics are universal and must be satisfied!

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Electron transfer: Moser-Dutton

Moser-Dutton semi-empirical electron transfer rate:

$$k_{ET}^{back} = k_{ET} \exp\left(\frac{\Delta G_0}{kT}\right) \Rightarrow \ln k_{ET}^{back} = \ln k_{ET} + \left(\frac{\Delta G_0}{kT}\right)$$

Note: In classical expression, thermodynamics is automatically satisfied:

$$\ln k_{ET}^{back} = \ln k_{ET} + \left(\frac{\Delta G_0}{kT}\right) = \text{const} - \left(\frac{2\sqrt{2m_e \Delta E_b}}{\hbar}\right) R_{DA} - \frac{[\Delta G_0 + \lambda]^2}{4\lambda kT} + \left(\frac{\Delta G_0}{kT}\right)$$

$$\begin{aligned} \text{Evaluate: } -\frac{[\Delta G_0 + \lambda]^2}{4\lambda kT} + \left(\frac{\Delta G_0}{kT}\right) &= -\frac{\Delta G_0^2 + 2G_0\lambda + \lambda^2}{4\lambda kT} + \frac{4\lambda \Delta G_0}{4\lambda kT} = \\ &= -\frac{\Delta G_0^2 - 2G_0\lambda + \lambda^2}{4\lambda kT} = -\frac{[-\Delta G_0 + \lambda]^2}{4\lambda kT} \end{aligned}$$

$$\ln k_{ET}^{back} = \text{const} - \left(\frac{2\sqrt{2m_e \Delta E_b}}{\hbar}\right) R_{DA} - \frac{[-\Delta G_0 + \lambda]^2}{4\lambda kT} \quad \gamma \equiv \frac{1}{4kT \ln 10} = 4.23 eV^{-1}$$

Note: the sign of ΔG_0 is opposite since donor and acceptor are swapped in back transfer.

However: γ has been adjusted in the semi-empirical formula (3.1eV instead of 4.23eV), which is inconsistent with thermodynamics for uphill ET!

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Electron transfer: Moser-Dutton

Moser-Dutton semi-empirical electron transfer rate:

$$\log_{10} k_{down}^{up} \approx 13 - 0.6 \left[\text{\AA}^{-1} \right] \left(R_{DA} - 3.6 \left[\text{\AA} \right] \right) - 3.1 \left[eV^{-1} \right] \frac{[\Delta G_0 + \lambda]^2}{\lambda}$$

$$k_{ET}^{uphill} = k_{ET}^{downhill} \exp\left(\frac{\Delta G_0}{kT}\right) \quad \text{Different equations for downhill and uphill}$$

ΔG_0 is assumed to be negative in both equations! I.e. this is for downhill.

These equations will not work for the inverted Marcus region, i.e., they only work for $-\Delta G_0 < \lambda$.

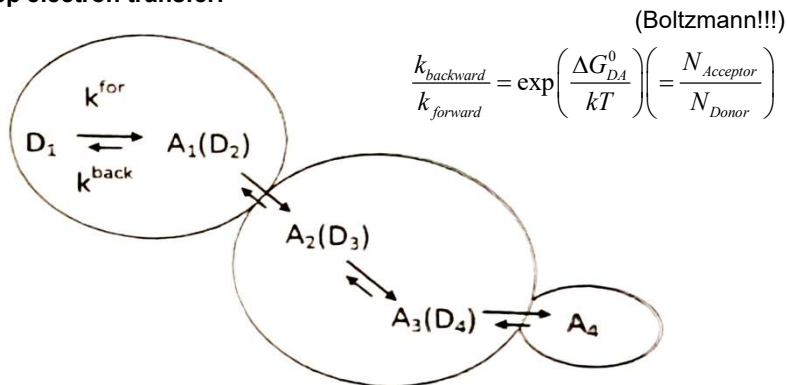
Typical values of $\lambda = 0.5 \dots 1.0$ eV, most commonly used $\lambda = 0.7$ eV.

In these equations, the distance is in $\text{\AA}$, and energy is in eV,

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Multistep Electron Transfer

Multistep electron transfer:



Goal: long-living charge-separated state to run slower chemical reactions

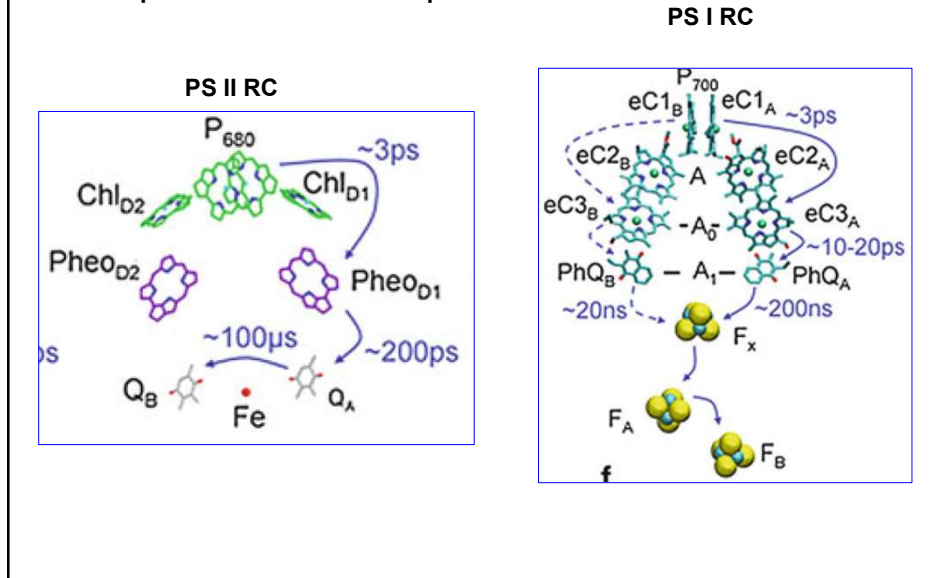
Achieved by:

- Multistep reaction, each step is consequently slower than the previous one
- Downhill transfer ($\Delta G^0 < 0$), backtransfer must be slower than forward transfer

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Electron transfer: examples

Multistep electron transfer: example



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Redox potential

Redox potential E_0 of an electron in a molecule is the difference in electric potential felt by the electron in that molecule and the potential felt by an electron in a hydrogen atom in water.

SI units: Volts

Redox potentials are different for *oxidation* and *reduction*.

Reduction – gaining an electron, the acceptor is being reduced

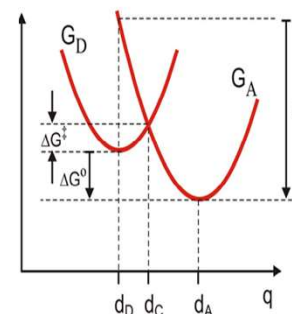
Oxidation – losing an electron, the donor is being oxidized

Example: electron on donor has higher energy than on acceptor $\rightarrow$ redox potential of donor is *more* negative than that of acceptor. Energy ΔG^0 , when measured in eV, is numerically the same as redox potential difference between acceptor and donor (but has opposite sign since charge e is negative):

$$\Delta G^0 = G_A(d_A) - G_D(d_D) \quad (<0)$$

$$\Delta G^0 = e\Delta E \quad \leftarrow \Delta E \text{ is redox potential difference}$$

$$\Delta E = E_D - E_A > 0 \text{ (in this example)}$$



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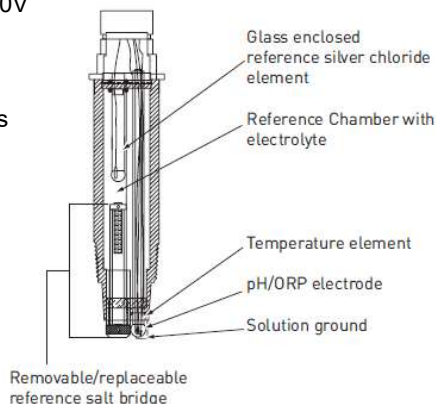
Redox potential

Measuring:

The absolute values are measured in respect to standard electrode.

If redox potential of your sample in solvent is the same, as in reference solvent, there will be 0V difference between electrodes

The difference in voltage is the difference in redox potentials of two solutions – one gives electrons away easier, than the other



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Redox potential

Calculating:

The redox potential is the difference between energies of neutral and ionized molecules – i.e., one can calculate energies of neutral and ionized molecules and find the difference using quantum mechanics (Qchem, Gaussian).

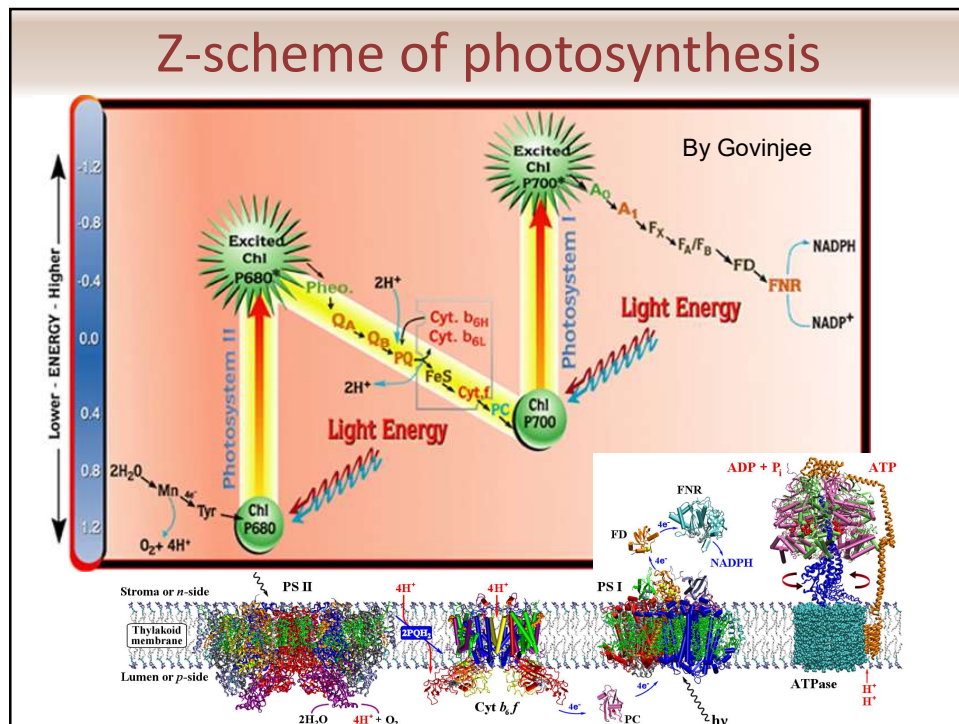
Note that to obtain *absolute* conventional energies, a standard Gibbs energy for $\text{H}^+ \rightarrow \frac{1}{2} \text{H}_2$ needs to be added to the difference, as that potential is taken to be zero. Free energy of H_2 reduction is:

$$\left(\Delta_r G^\circ \left(\text{H}^+ \rightarrow \frac{1}{2} \text{H}_2 \right) \right) = 4.281 \text{ eV}$$

Note: since redox reactions always involve a change in the charge of a molecule, the environment can have a huge effect on the redox potential, as it can greatly reduce the energy difference due to screening.

For more on H_2 reduction potential and calculations see also:
Tazhigulov et. al, Phys Chem Chem Phys, 2019 (DOI: 10.1039/c9cp01533g)

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